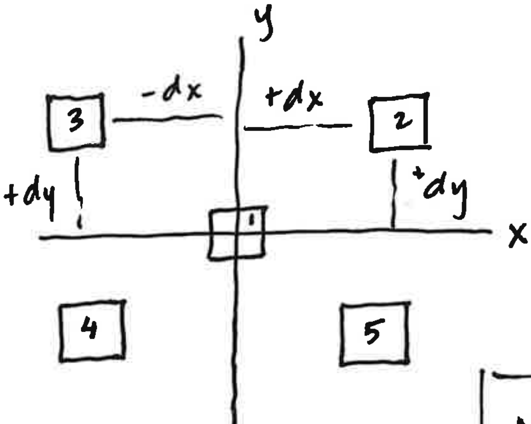
Interesting Product of Inertia (POI)

Example Problem 1



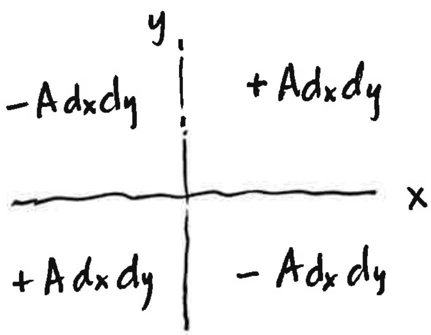
Five different Areas, located differently in x-y plane.

Determine I_{xy} for each area.

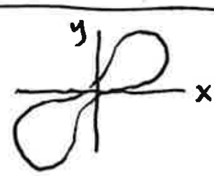
PAT: $I_{xy} = \bar{I}_{x'y'} + A \cdot d_x d_y$, with $\bar{I}_{x'y'} = 0$

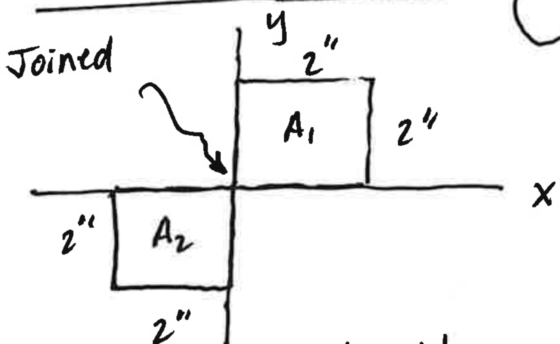
No	A	$\bar{I}$	d_x	d_y	I_{xy}
1	A	0	0	0	0
2	A	0	+	+	+
3	A	0	−	+	−
4	A	0	−	−	+
5	A	0	+	−	−

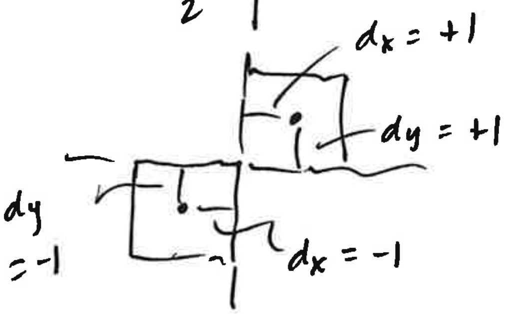
$= \boxed{A d_x d_y}$



Ex Problem 2







I_{xy} for this area:

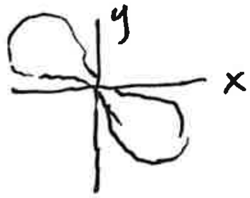
$$I_{xy} = \Sigma(\bar{I} + A d_x d_y)$$

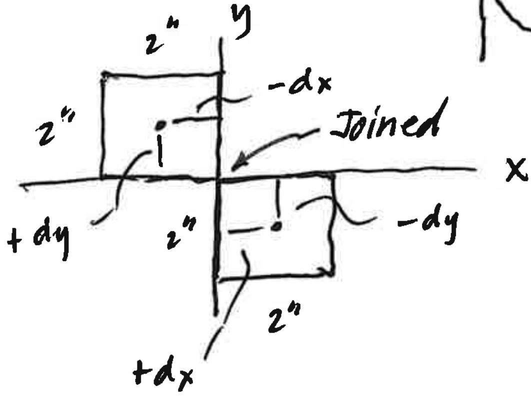
$$= \left[0 + (2)(2)(+1)(+1)\right]$$

$$+ \left[0 + (2)(2)(-1)(-1)\right]$$

$$I_{xy} = 4 + 4 = \boxed{+8 \text{ in}^4 = I_{xy}}$$

Ex Prob 3





$$I_{xy} = \Sigma\left(\bar{I} + A d_x d_y\right)$$

$$= \left[0 + (2)(2)(-1)(+1)\right]$$

$$+ \left[0 + (2)(2)(+1)(-1)\right]$$

$$I_{xy} = -4 - 4 = \boxed{-8 \text{ in}^4 = I_{xy}}$$